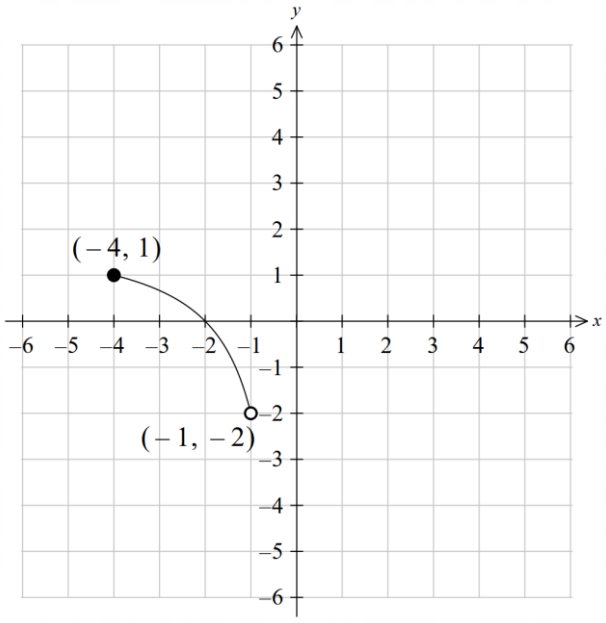
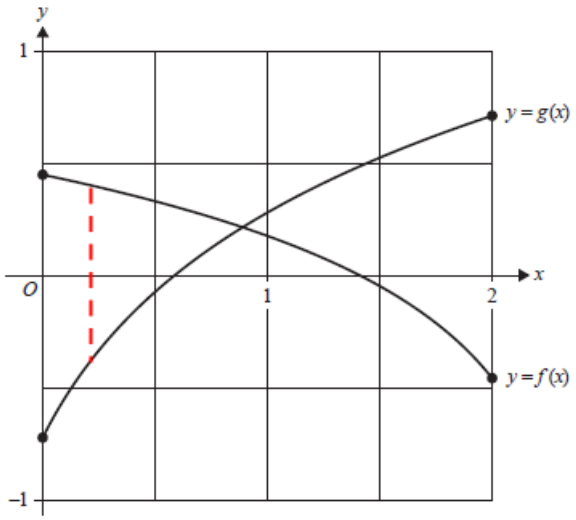
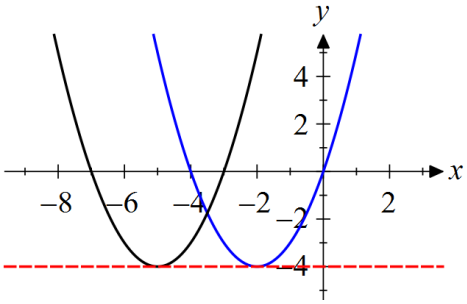
2026 VCE Mathematical Methods 2 (NHT) external assessment report

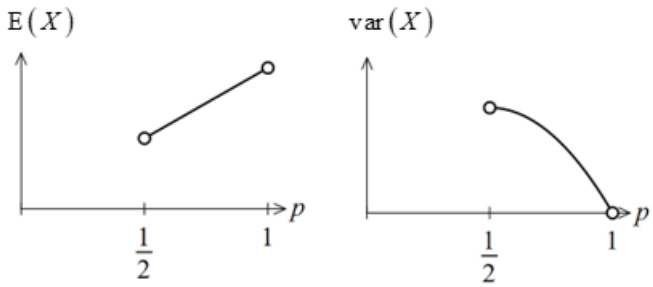
This report provides sample answers, or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

Section A

<table>
<tr><th>Question</th><th>Correct answer</th><th>Comments</th></tr>
<tr><td>1</td><td>B</td><td>$(0.719, 0.815)$</td></tr>
<tr><td>2</td><td>A</td><td>The table below shows the values of i and j for the first 9 iterations of the algorithm.
<table>
<tr><th>i</th><th>j</th></tr>
<tr><td>1</td><td>1</td></tr>
<tr><td>1</td><td>2</td></tr>
<tr><td>1</td><td>3</td></tr>
<tr><td>1</td><td>4</td></tr>
<tr><td>1</td><td>5</td></tr>
<tr><td>1</td><td>6</td></tr>
<tr><td>2</td><td>1</td></tr>
<tr><td>2</td><td>2</td></tr>
<tr><td>2</td><td>3</td></tr>
</table>
The ninth line printed is 2, 3.</td></tr>
<tr><td>3</td><td>D</td><td>$\Pr(X > 20) = \Pr\left(Z > \frac{20-15}{\sqrt{4}}\right)$
$= \Pr(Z > 2.5)$
$= 1 - \Pr(Z < 2.5)$</td></tr>
</table>

Question	Correct answer	Comments
4	A	$f(x)=\frac{4}{x}+2$ Solve $\frac{4}{x}+2=1$, $x=-4$ and $\frac{4}{x}+2=-2$, $x=-1$.  If $-4 \le x < -1$, then the range of $f(x)$ is $-2 < y \le 1$.
5	D	$f(x)=e^{-x}-2$, $f'(x)=-e^{-x}$ $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ $= a - \frac{e^{-a}-2}{-e^{-a}}$ $= a+1-2e^{a}$
6	A	$y=2^{x-1}+3$, inverse swap x and y $x=2^{y-1}+3$ $x-3=2^{y-1}$ $\log_2(x-3)=y-1$ $y=\log_2(x-3)+1$ $y=\log_2(x-3)+\log_2(2)$ $y=\log_2(2x-6)$

Question	Correct answer	Comments
7	A	The equation $f(x)+g(x)=0$ is equivalent to $f(x)=-g(x)$. We are looking for a value of x such that $f(x)$ and $g(x)$ are of opposite sign but equal distance from the x-axis. The required x-value is approximately $x=0.2$, as shown on the graph below.  So, x is in the interval $\left[0,\frac{1}{2}\right)$.
8	C	The **first** step containing an error is Step 4. Step 4 – expressing $x=\frac{\log_2(3)}{\log_2\left(\frac{3}{4}\right)}$ as $x=\log_2\left(3-\frac{3}{4}\right)$ is **not** correct. Step 5 – expressing $x=\log_2\left(\frac{9}{4}\right)$ as $x=\sqrt{\frac{9}{4}}$ is **not** correct.
9	B	The most efficient method to solve this is by considering the graphs of $y=x^2+4x$ and $y=(x+5)^2-4$.  Both graphs have a turning point at $y=-4$, and this horizontal line is the only line that is tangent to both graphs. The gradient is zero.

Question	Correct answer	Comments
		Alternatively, students can consider the equations $x^2+4x=mx+c$ $x^2+(4-m)x-c=0$ and $(x+5)^2-4=mx+c$ $x^2+(10-m)x+(21-c)=0$ Both quadratic equations must have one solution, which means they must both have a discriminant of zero. $(4-m)^2+4c=0$ $(10-m)^2-4(21-c)=0$ Solving these equations gives $m=0$ and $c=-4$.
10	B	The mean and variance are given by $\mathrm{E}(X)=np$ and $\mathrm{var}(X)=np(1-p)$ We can consider these as functions of p, for a given value of n. The mean is a linear function of p, while the variance is a quadratic function of p with a local maximum at $p=\frac{1}{2}$.  As p increases, $\mathrm{E}(X)$ increases and $\mathrm{var}(X)$ decreases.
11	D	The graph of $y=g(x)$ has 2 x-intercepts, which are at $x=a$ and $x=b$. Substituting these into the equation $f(x)=0$ gives the equations $a^2+ac+ab=0$ and $b^2+bc+ab=0$ Solving these simultaneously gives $c=-(a+b)$. OR $f(x)=x^2+cx+ab=(x-b)(x-a)=x^2-(a+b)x+ab$ Equating coefficients $c=-(a+b)$.

Question	Correct answer	Comments
12	B	First notice that there are only 3 values of $x \in [0, 2\pi]$ where $f_1(x) = \sin(x) = 0$, which are 0, π and 2π. Now notice that for any integer n, $f_n(x) = \sin(nx) = 0$ when $x = 0, \pi$ or 2π. So exactly 3 values of $x \in [0, 2\pi]$ exist such that $f_1(x) = f_2(x) = f_3(x) = \ldots = f_{10}(x) = 0$.
13	C	The general solution (which can be found by hand or using CAS) is: $x = \frac{\pi}{6} + k\pi$ or $x = \frac{\pi}{3} + k\pi$, where $k \in Z$ Looking at each option: $\left\{\frac{-2\pi}{3} + k\pi,\ k \in Z\right\}$ is equivalent to $\left\{\frac{\pi}{3} + k\pi,\ k \in Z\right\}$, so it is a subset of the general solution. $\left\{\frac{\pi}{3} + 2k\pi,\ k \in Z\right\}$ is a subset of $\left\{\frac{\pi}{3} + k\pi,\ k \in Z\right\}$, so it is a subset of the general solution. $\left\{\frac{5\pi}{6} + k\pi,\ k \in Z\right\}$ is **not** a subset of the general solution (notice $\frac{5\pi}{6}$ is not a solution). $\left\{\frac{7\pi}{6} + 2k\pi,\ k \in Z\right\}$ is a subset of $\left\{\frac{\pi}{6} + k\pi,\ k \in Z\right\}$, so it is a subset of the general solution.
14	A	Given that the system of 2 equations has no solutions, the 2 lines must be parallel so their gradients must be equal. $\frac{b}{a} = -\frac{3}{2}$ Given that $x = 1$ and $y = 2$ is a solution to the second equation, $a(1) + b(2) = 2$ Solving these 2 equations for a and b gives $a = -1$ and $b = \frac{3}{2}$. Equation 2 becomes $-x + \frac{3y}{2} = 2$. Substituting $x = 4$ gives $y = 4$, so Option A is correct. (None of the other 3 pairs of x and y values satisfy Equation 2.)

Question	Correct answer	Comments
15	D	The graph of $y = g(x)$ is obtained by performing the following sequence of transformations to the graph of $y = f(x)$. Dilation by factor m from the x-axis Dilation by factor $\frac{1}{n}$ from the y-axis Translation by c units up Transformations 1 and 3 transform the average value from 2 to $2m + c$. Transformation 2 transforms the interval from $[-1,1]$ to $\left[\frac{-1}{n}, \frac{1}{n}\right]$.
16	B	The period of $f(x)$ is 12. Of the 3 transformations, only the dilation by a factor of $\frac{1}{2}$ from the y-axis affects the period. $\text{period}_h = \frac{1}{2} \times 12$ $= 6$
17	D	Notice that $g(0) = 0$ and $f(0) = 1$, so $(f \circ g)(0) = 1$. Notice that $g(0.5) = 0$ and $f(0) = 1$, so $(f \circ g)(0.5) = 1$. Finally, notice that $g(0.25) = g(0.75) = 1$ and $f(1) = 0$, so $(f \circ g)(0.25) = (f \circ g)(0.75) = 0$. Only Option D passes through all of these points. Alternatively, f is a strictly decreasing function, so $(f \circ g)(x)$ must be decreasing when g is increasing and vice versa. Only Option D shows this behaviour.
18	C	Notice that $f'(3) = 0$, so the graph of $y = f(x)$ must have a stationary point at $x = 3$. Notice that $f'(x) > 0$ for all $x \neq 3$, so the graph of $y = f(x)$ should be strictly increasing. Only Option C meets both conditions.

Question	Correct answer	Comments
19	C	Let $X \sim \text{Bi}(3, p)$. The probability the player wins the game is: $\Pr(X = 2) + \Pr(X = 3)$ $= \binom{3}{2} p^2(1-p) + p^3$ $= 3p^2 - 2p^3$ The probability the player wins the game, but misses one of their first 2 attempts is: $p(1-p)p + (1-p)p^2$ $= 2p^2(1-p)$ So the required conditional probability is: $\frac{2p^2(1-p)}{3p^2 - 2p^3}$ $= \frac{2(1-p)}{3-2p}$
20	D	The surface area is $A = 2\pi r^2 + 2\pi rh$. Using the volume $V = \pi r^2 h$, we can obtain $h = \frac{V}{\pi r^2}$, then write the surface area as: $A = 2\pi r^2 + 2\pi r\left(\frac{V}{\pi r^2}\right)$ $= 2\pi r^2 + \frac{2V}{r}$ This function has a minimum when: $\frac{dA}{dr} = 0$ $4\pi r - \frac{2V}{r^2} = 0$ $r = \sqrt[3]{\frac{V}{2\pi}}$ Substituting this back into the surface area function gives the minimum possible area of $3\left(\sqrt[3]{2\pi V^2}\right)$ cm^2.

Section B

Question 1a.

$f(0) = 33\,°\text{C}$

Question 1b.

$y = 6$

Question 1c.

$k = \ln(3)$

$f(t) = 6 + 27e^{-\ln(3)t}$

$= 6 + 27e^{\ln\left(3^{-t}\right)}$

$= 6 + 27\left(3^{-t}\right)$

$= 6 + 3^{3-t}$

Question 1d.

Solve $f(t) = 8.5$

$t = 2.17\text{ h}$

Question 1ei.

$g(t) = \int 14e^{-t}\,dt$

$g(t) = -14e^{-t} + c$

$g(3) = f(3) = 7$

$g(t) = -14e^{-t} + 14e^{-3} + 7$

Question 1eii.

$$h'(t)=\begin{cases}-27\ln(3)3^{-t} & 0\leq t<3\\ 14e^{-t} & t>3\end{cases}$$

The left derivative of $h(t)$ at $t=3$ is:

$$-27\ln(3)3^{-3}=-\ln(3)$$

While the right derivative of $h(t)$ is:

$$14e^{-3}\neq-\ln(3)$$

$h(t)$ is not differentiable at $t=3$.

Question 1fi.

1. Dilation by a factor of $\frac{9}{5}$ from t-axis
2. Translation 32 units up

Question 1fii.

$$\frac{9}{5}p(t)+32=\frac{9}{5}\left(p(t)+\frac{160}{9}\right)$$

1. Translation $\frac{160}{9}$ units up
2. Dilation by a factor of $\frac{9}{5}$ from t-axis

Question 2a.

$(-1,-0.5)$ and $(1,0.5)$

Question 2b.

$$-\int_{-1}^{0}f(x)dx+\int_{0}^{1}f(x)dx=2\int_{0}^{1}f(x)dx$$

$$=\ln(2)$$

Question 2c.

At $x=0$ the equation of the tangent line is $y=x$.

Question 2d.

Solve $f'(x)=\frac{-1}{8}$, $x=\pm\sqrt{3}$

$y=-\frac{1}{8}x-\frac{3\sqrt{3}}{8}$ or $y=-\frac{1}{8}x+\frac{3\sqrt{3}}{8}$

Question 2ei.

$$\frac{f(b)-f(a)}{b-a}=\frac{1}{b-a}\left(\frac{b}{b^2+1}-\frac{a}{a^2+1}\right)$$

$$=\frac{1}{b-a}\cdot\frac{b\left(a^2+1\right)-a\left(b^2+1\right)}{\left(a^2+1\right)\left(b^2+1\right)}$$

$$=\frac{1}{b-a}\cdot\frac{a^2b+b-ab^2-a}{\left(a^2+1\right)\left(b^2+1\right)}$$

$$=\frac{1}{b-a}\cdot\frac{ab(a-b)+(b-a)}{\left(a^2+1\right)\left(b^2+1\right)}$$

$$=\frac{1}{b-a}\cdot\frac{(b-a)(1-ab)}{\left(a^2+1\right)\left(b^2+1\right)}$$

$$=\frac{1-ab}{\left(a^2+1\right)\left(b^2+1\right)}$$

Question 2eii.

Method 1 ('hence')

For $f(a)=f(b)$, the average rate of change must be zero.

This occurs only when the numerator is zero.

$1-ab=0$

$a=\frac{1}{b}$

Method 2 ('otherwise')

Solving $f(a)=f(b)$ with $a<b$ gives $a=\frac{1}{b}$.

Question 2fi.

Solve $g'(x) = 0$ for x.

$x = \pm\sqrt{k}$

Question 2fii.

For g to be strictly decreasing for all $x \geq 1$, we require $g(x) < g(1)$ for all $x > 1$.

Therefore, g cannot have a turning point where $x > 1$.

Observing that the graph of $y = g(x)$ has a local minimum at $x = -\sqrt{k}$ and a local maximum at $x = \sqrt{k}$, we require $\sqrt{k} \leq 1$.

$0 < k \leq 1$

Question 2fiii.

Stationary points are $\left(\sqrt{k}, \frac{1}{2\sqrt{k}}\right)$ and $\left(-\sqrt{k}, -\frac{1}{2\sqrt{k}}\right)$.

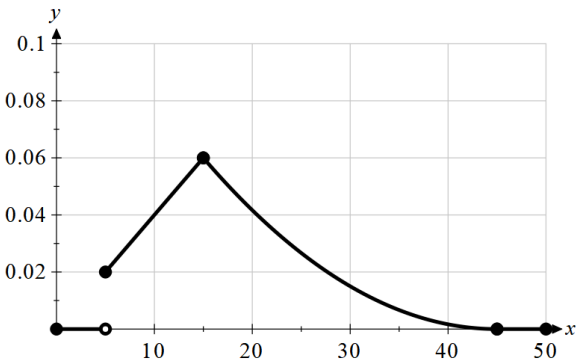
The line L through these points has equation $y = \frac{x}{2k}$.

$$\text{Area} = \int_{-\sqrt{k}}^{0} \frac{x}{2k} - g(x)\,dx + \int_{0}^{\sqrt{k}} g(x) - \frac{x}{2k}\,dx = 2\int_{0}^{\sqrt{k}} g(x) - \frac{x}{2k}\,dx$$

$$= \ln(2) - \frac{1}{2} = \frac{2\ln(2) - 1}{2}$$

Question 3ai.

Sketch the graph $y = f(x)$ for $0 \leq x \leq 50$.



Question 3aii.

$$\mathrm{E}(X) = \int_5^{45} x \cdot f(x)dx = \frac{107}{6}$$

$$= 17.8 \text{ km}$$

Question 3bi.

$$Y \sim \mathrm{N}(16.5, 5^2)$$

$$\Pr(Y > 12) = 0.816$$

Question 3bii.

$\Pr(Y > y_{\min}) = 0.15$ or $\Pr(Y \leq y_{\min}) = 0.85$

Use inverse normal to get $y_{\min} = 21.7$ km.

Question 3biii.

$W \sim \mathrm{Bi}(10, 0.15)$ or $\binom{10}{2}(0.15)^2(0.85)^8$

$$\Pr(W = 2) = 0.276$$

Question 3biv.

$$\Pr\left(\hat{P} > \frac{1}{3}\right) = \Pr\left(W > \frac{10}{3}\right) = \Pr(W \geq 4)$$

$$= 0.050$$

Question 3bv.

$$\hat{P} \sim \text{N}\left(0.15, \frac{0.15 \times 0.85}{n}\right)$$

$$\Pr\left(\hat{P} > 0.1\right) = \Pr\left(Z > \frac{0.1 - \text{E}\left(\hat{P}\right)}{\text{sd}\left(\hat{P}\right)}\right)$$

$$\frac{0.1 - E(\hat{P})}{\text{sd}(\hat{P})} < -1.644...$$

$$\frac{0.1 - 0.15}{\sqrt{\frac{0.15 \times 0.85}{n}}} < -1.644...$$

$$n > 137.98$$

$$n = 138$$

Question 3ci.

$$\Pr(\text{dist} > 15)$$

$$= \frac{1}{2} \times \Pr(X > 15) + \frac{1}{2} \times \Pr(Y > 15)$$

$$= \frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \times 0.6179...$$

$$\approx 0.609$$

Question 3cii.

$$\Pr(\text{app A} \mid \text{dist} > 15)$$

$$= \frac{\Pr(\text{app A} \cap \text{dist} > 15)}{\Pr(\text{dist} > 15)}$$

$$= \frac{\frac{1}{2} \times \Pr(X > 15)}{\frac{1}{2} \times \Pr(X > 15) + \frac{1}{2} \times \Pr(Y > 15)}$$

$$= \frac{\frac{3}{5}}{\frac{3}{5} + 0.6179...}$$

$$= \frac{0.3}{0.6089...}$$

$$\approx 0.493$$

Question 4a.

$f'(x) = 0$

$x = 1, 2, 3, 4$

Question 4bi.

$k = 0, 3$

Question 4bii.

$k \in (0, 1) \cup (2, 3)$

Question 4ci.

$(2, 0)$ and $(4, -1)$

Question 4cii.

Translate the graph of $f(x)$, 5 units left and then 3 units down.

$b = 5$ and $c = -3$

Question 4di.

$x \in [0, 2]$

Question 4dii.

When $f(1) = 2, f^{-1}(2) = 1$

$h_1(2) = 1$

Question 4ei.

Because f has turning points at $x = 1, 2, 3, 4.$

The restricted function must be one-to-one, so it cannot contain a turning point (except at an endpoint).

Question 4eii.

$m = 3, 4$

Question 4fi.

$\text{ran}_g = [2, 3.792) = \left[2, \frac{474}{125}\right)$

$\text{dom}_f = [0, 5]$

$\text{ran}_g \subset \text{dom}_f$

$f(g(x))$ is defined.

Question 4fii.

For $x \in [2, 3.792)$, f has a minimum of $f(2) = 0.$

So, the minimum value of $f(g(x))$ is 0.

It occurs when $g(x) = 2$ (i.e. when $x = 1$).

Question 4fiii.

Using the chain rule:

$f'(g(x))g'(x)=0$, $f'(g(x))=0$ or $g'(x)=0$

If $g'(x)=0$

$x=\frac{-1}{3}$ or $x=1$

If $f'(g(x))=0$

$g(x)=2$ or $g(x)=3$

$x=0,\ 1,\ \frac{1\pm\sqrt{5}}{2}$

$x=\frac{-1}{3},\ 0,\ 1,\ \frac{1\pm\sqrt{5}}{2}$